Horizon-Specific Causal Graph Generation and Horizon-Indexed Causal Propagation (HGP)

Architecture and working of the proposed model | LargeST-GBA 2021, 5-minute flow, $N = 2{,}056$ sensors | 12 November 2026

> **Status.** Part A (graph generation) is implemented; its knockoff stage has been run at $h = 1$ only, so G_3, G_6, G_{12} from the knockoff filter are still to be computed (the VAR impulse-response graphs exist at every horizon). Part B (HICP) is a design specification and has not been built or trained. No accuracy claim is made in this document.

0. Notation and data

Symbol	Meaning
$N = 2{,}056$	sensors (nodes); $T_{\text{tr}} = 84{,}096$ training rows (5-minute steps); validation and test follow chronologically
$x_t \in \mathbb{R}^N$, $X \in \mathbb{R}^{T_{\text{tr}} \times N}$	flow at step t; the training flow matrix (observed values only; missing entries are masked, never filled, in Part A)
$\mathcal{H} = \{3, 6, 12\}$	forecast horizons (15, 30, 60 minutes); $h = 1$ is kept only for comparison with earlier runs
E_{road}, $\mathcal{N}_i$	road-adjacency edge set; road neighbours of sensor i
$G_h \in \mathbb{R}^{N \times N}$	generated graph for horizon h; convention $G_h[i, j]$ = weight of the directed edge *source* $j \to$ *target* i; zero diagonal
$T = 12$, F, $d = 64$	input window length, input features per node per step, hidden width
B	batch size

Part A — horizon-specific causal graph generation

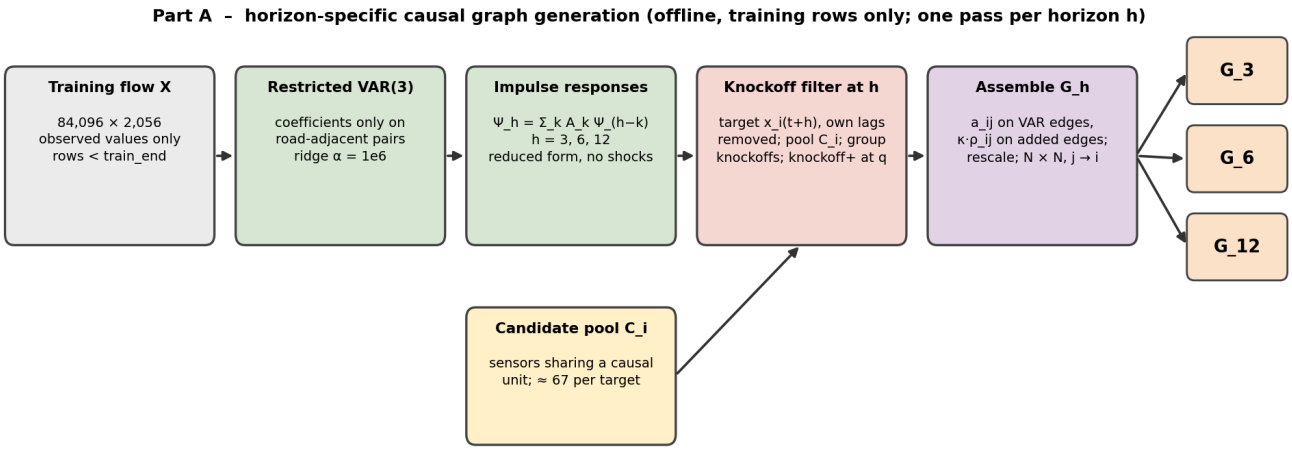


Figure 1. Part A. One pass per horizon h: the same fitted VAR gives Ψ_h; the knockoff filter is run on the h-step-ahead relation; the output is one directed $N \times N$ graph per horizon. Everything uses training rows only.

A.1 Restricted vector autoregression

A VAR of order $p = 3$ is fitted with coefficients permitted only on road-adjacent pairs and the diagonal:

$$x_t = \sum_{k=1}^{p} A_k \, x_{t-k} + \varepsilon_t, \qquad (A_k)_{ij} = 0 \ \text{ unless } (i, j) \in E_{\text{road}} \text{ or } i = j. \tag{1}$$

Each equation i is estimated separately by ridge regression on its allowed regressors (observed rows only):

$$\hat{a}_i = \arg\min_{a_i} \sum_t \Big(x_{i,t} - \sum_{k=1}^{p} \sum_{j \in \mathcal{N}_i \cup \{i\}} a_{ij}^{(k)}\, x_{j,t-k} \Big)^2 + \lambda \|a_i\|_2^2, \qquad \lambda = 10^6. \tag{2}$$

An unrestricted VAR on $2{,}056$ sensors would have about 12.7 million coefficients per lag and is ill-posed with 84k rows; the restriction makes the fit feasible. The ridge shrinks $\hat{A}_k$ towards zero, so the magnitudes of Ψ_h are biased downward; rankings and supports are what the pipeline uses.

A.2 Impulse responses per horizon

The h-step impulse responses follow the standard recursion

$$\Psi_0 = I, \qquad \Psi_s = \sum_{k=1}^{\min(s,p)} A_k\, \Psi_{s-k}, \quad s = 1, 2, \dots \tag{3}$$

so that $\Psi_h[i,j] = \partial\, \mathbb{E}[x_{i,t+h}] \,/\, \partial \varepsilon_{j,t}$ is the *reduced-form* effect of a unit innovation at j on i after h steps. No shock identification (no Cholesky ordering, no structural restriction) is imposed; the quantity is a multi-step *predictive* effect and is described as Granger-type, not structural, causality. The standardised effect used for ranking and weighting is

$$a_{ij}^{h} = |\Psi_h[i,j]|\, \frac{\sigma_j}{\sigma_i}, \qquad \sigma_j = \text{training standard deviation of sensor } j. \tag{4}$$

Because the A_k live on the road support, $\operatorname{supp}(\Psi_h) \subseteq \operatorname{supp}\big((I + A_{\text{road}})^h\big)$: the horizon-$h$ response can only reach sensors within h road hops (measured: 26, 101, 231 and 524 reachable sources per target at $h = 1, 3, 6, 12$). Non-road dependencies enter only through the candidate pool of A.3.

A.3 Candidate pool

For target i the candidate set $\mathcal{C}_i$ is the set of sensors that share a causal unit with i (mean $|\mathcal{C}_i| = 67$, range 0–173); about 63% of pool members lie outside the VAR support. $\mathcal{C}_i$ is the only place where an edge that the road-restricted VAR cannot represent can be proposed.

A.4 Knockoff selection at horizon h

For each target i and horizon h the following steps are carried out on the training rows.

(i) Pre-processing. Time-of-day means are removed, $\tilde{x}_{j,t} = x_{j,t} - m_j(\text{tod}(t))$, and only every third row is used (15-minute spacing) to weaken serial dependence.

(ii) Response with own history removed. The h-step-ahead target is residualised on its own last $L_y = 3$ values, so the statistic measures what sources add beyond the target's own past:

$$r_i(t) = \tilde{x}_{i,t+h} - \hat{\mathbb{E}}\big[\tilde{x}_{i,t+h} \,\big|\, \tilde{x}_{i,t}, \tilde{x}_{i,t-1}, \tilde{x}_{i,t-2}\big]. \tag{5}$$

(iii) Design matrix. $D_i \in \mathbb{R}^{n imes |\mathcal{C}_i|}$ holds one column per candidate $j \in \mathcal{C}_i$: its value $ildex_{j,t}$ at the forecast origin, residualised on the target's own lags exactly as r_i is, then standardised. Only the target carries lags; each candidate enters through its single value h steps before the target.

(iv) Groups. Candidates are clustered (average linkage) so that members of a group have $|\text{corr}| \geq 0.5$; let $\mathcal{G}_i$ be the resulting groups. Grouping is necessary because highly collinear sensors would otherwise mask each other.

(v) Group knockoffs. With $\Sigma = \text{Cov}(D_i)$, second-order group knockoffs $\tilde{D}_i$ are sampled so that

$$\text{Cov}\begin{pmatrix} D_i \\ \tilde{D}_i \end{pmatrix} = \begin{pmatrix} \Sigma & \Sigma - S \\ \Sigma - S & \Sigma \end{pmatrix}, \qquad S = \gamma\ \text{blockdiag}\,(\Sigma_g)_{g \in \mathcal{G}_i}, \tag{6}$$

with $\gamma \in (0, 1]$ the largest value keeping $2\Sigma - S \succeq 0$ (measured $\gamma \approx 0.19$ with groups, ≈ 0.05 without). The knockoffs copy the within- and between-group correlation structure of the real candidates but carry no information about r_i beyond what D_i carries.

(vi) Statistic. A lasso path of r_i on $[D_i \ \tilde{D}_i]$ gives, for each column c, the penalty $\lambda_c = \sup\{\lambda : \hat{\beta}_c(\lambda) \neq 0\}$ at which it enters. The group statistic is the entry difference between a group and its knockoff copy,

$$W_g = \max_{c \in g} \lambda_c \ - \ \max_{c \in g} \tilde{\lambda}_c, \qquad g \in \mathcal{G}_i. \tag{7}$$

Large positive W_g means the real group entered the model well before its twin; a null group is equally likely to give $\pm W_g$.

(vii) Knockoff+ threshold and selection.

$$\tau_q = \min\left\{t > 0 : \frac{1 + \#\{g : W_g \leq -t\}}{\max\left(1, \ \#\{g : W_g \geq t\}\right)} \leq q\right\}, \qquad \hat{\mathcal{S}}_i^h = \{g : W_g \geq \tau_q\}. \tag{8}$$

One representative per selected group is kept (the member with the largest individual statistic), and at most 25 sources per target are retained. Targets whose groups all fall below τ_q receive no sources.

Guarantee and its limits. Under the model-X assumptions (the candidate distribution is known and rows are exchangeable) the knockoff+ procedure satisfies

$$\mathrm{FDR} = \mathbb{E}\left[\frac{\#\{\text{selected null groups}\}}{\max(1, \ \#\{\text{selected groups}\})}\right] \leq q. \tag{9}$$

On traffic data both assumptions hold only approximately: the covariate law is replaced by its second moments (Gaussian second-order knockoffs), and rows are serially dependent despite the detrending, own-lag residualisation and 15-minute thinning. The guarantee is therefore stated as approximate and checked empirically by a placebo test (the candidate block is shifted by a random whole number of days so that no true edge exists; under this global null the share of targets with any selection must be at most q). At $h = 1$ the placebo rate was at the nominal level. The choice of q is a power/FDR trade-off made on validation data: on this network $q = 0.05$ selects 867 edges in total (2% of targets have any source), $q = 0.2$ gives 21,593 and $q = 0.3$ gives 25,451 edges (91% of targets).

A.5 Graph assembly

Let $\hat{\mathcal{S}}_i^h$ now denote the set of representative source sensors for target i. Edge weights are

$$w_{ij}^h = \begin{cases} a_{ij}^h, & \Psi_h[i, j] \neq 0 \quad \text{(VAR-supported edge)} \\ \kappa_h \, \rho_{ij}^h, & \text{otherwise} \quad \text{(edge outside the VAR support)} \end{cases}, \qquad j \in \hat{\mathcal{S}}_i^h, \tag{10}$$

where ρ_{ij}^h is the partial correlation of the h-step forecasting relation (shrunk inverse covariance) and κ_h is one global constant per horizon, $\kappa_h = \mathrm{median}\{a_{ij}^h : \text{VAR edges}\} \,/\, \mathrm{median}\{\rho_{ij}^h : \text{added edges}\}$, so that the two weight families sit on one scale. All weights are then multiplied by one constant c_h so that the mean non-zero weight equals that of the VAR-only graph at the same horizon (this keeps the self-to-neighbour balance comparable across variants):

$$G_h[i, j] = c_h \, w_{ij}^h \ \ (j \in \hat{\mathcal{S}}_i^h), \qquad G_h[i, j] = 0 \ \text{otherwise}, \qquad G_h[i, i] = 0. \tag{11}$$

At $h = 1$: 25,451 edges, of which 9,935 are VAR-supported and 15,516 added; 12.4 sources per target; maximum weight 0.56.

A.6 Are the graphs structurally different across horizons?

Weights alone do not establish that G_3, G_6, G_{12} differ. The following are reported per pair of horizons (h, h'): edge-set Jaccard overlap $J = |E_h \cap E_{h'}| \,/\, |E_h \cup E_{h'}|$; density; in- and out-degree distributions;

Spearman rank correlation of weights on the common support; sign-flip rate of Ψ_h against $\Psi_{h'}$; and bootstrap stability of each E_h across training sub-periods. Measured so far on the VAR top-25 graphs: $J(G_1, G_h) = 0.53, 0.48, 0.41$ for $h = 3, 6, 12$; 37–48% of the edges at $h \geq 3$ are absent at $h = 1$; 23–30% of shared entries change sign; the per-target correlation of $\log a^1$ with $\log a^6$ is 0.42 (a pure rescaling would give 1.0). Caveat: part of the change is the growing road-hop reach, so a road-graph-power baseline (A_{road}^k, no dynamics) is included among the controls.

Part B — HICP: horizon-indexed causal propagation

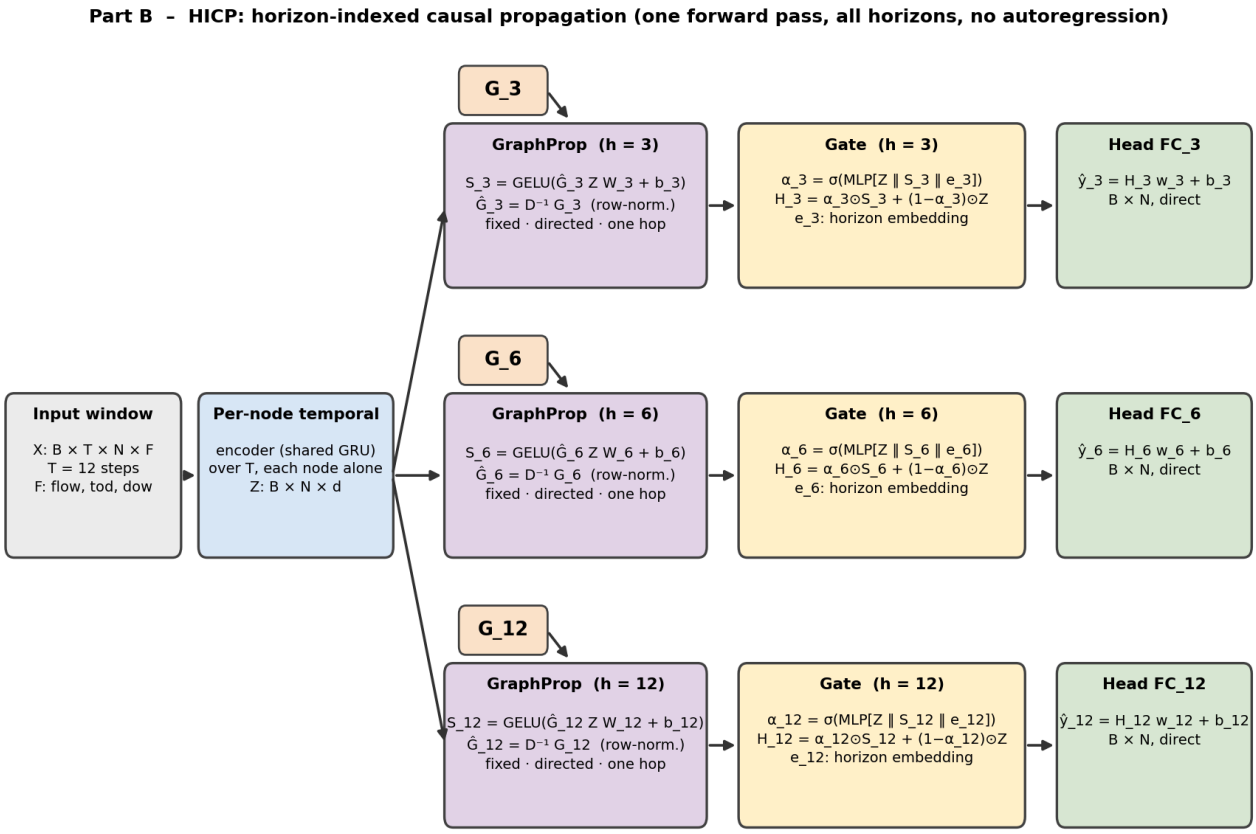


Figure 2. HICP. A shared per-node temporal encoder produces Z; each horizon has its own propagation branch over its own fixed graph G_h, a horizon-conditioned gate, and a prediction head. All horizons are produced in one forward pass. Between experimental configurations only the adjacency tensors change.

B.1 Input

$X \in \mathbb{R}^{B \times T \times N \times F}$ with $T = 12$ steps. Per step and node the F features are the standardised flow (training mean and standard deviation per sensor; gaps in model inputs filled by the existing gap-band policy and flagged), $\sin$ and $\cos$ of time-of-day, and a day-of-week indicator. Targets are the standardised flows at $t + h$, $h \in \mathcal{H}$, with a mask M_h marking observed entries.

B.2 Per-node temporal encoder (shared weights)

Each node is encoded independently by a GRU whose parameters are shared across nodes, so the parameter count does not grow with N:

$$u_{n,t} = \sigma\big(W_u x_{n,t} + U_u \mathbf{h}_{n,t-1} + b_u\big), \qquad r_{n,t} = \sigma\big(W_r x_{n,t} + U_r \mathbf{h}_{n,t-1} + b_r\big),$$
$$\tilde{\mathbf{h}}_{n,t} = \tanh\big(W_c x_{n,t} + U_c(r_{n,t} \odot \mathbf{h}_{n,t-1}) + b_c\big), \qquad \mathbf{h}_{n,t} = u_{n,t} \odot \mathbf{h}_{n,t-1} + (1 - u_{n,t}) \odot \tilde{\mathbf{h}}_{n,t}, \tag{12}$$

$$Z_n = \mathbf{h}_{n,T} \in \mathbb{R}^d, \qquad Z \in \mathbb{R}^{B \times N \times d}. \tag{13}$$

Z contains no cross-sensor information. With all gates closed (B.4) the model reduces to a per-node forecaster, which is the strongest configuration observed so far at 60 minutes in the A3T-GCN experiments; it is therefore the floor that any graph contribution must improve on.

B.3 Graph normalisation and horizon-specific propagation

Each generated graph is row-normalised over its incoming edges (in-degree normalisation, which respects edge direction; the symmetric $D^{-1/2}(A+I)D^{-1/2}$ used by GCN-style models mixes out-degree into a directed graph and is not used here). No self-loop is added because the self path is carried by Z:

$$\hat{G}_h = D_h^{-1} G_h, \qquad D_h = \mathrm{diag}\Big(\max\big(\textstyle\sum_j G_h[i,j],\ \epsilon\big)\Big). \tag{14}$$

The horizon-h spatial representation is one directed hop over $\hat{G}_h$ followed by a linear map and a non-linearity, applied to the encoded source histories:

$$S_h = \phi\big(\hat{G}_h\, Z\, W_h + b_h\big) \in \mathbb{R}^{B\times N\times d}, \qquad (\hat{G}_h Z)_i = \sum_j \hat{G}_h[i,j]\, Z_j, \tag{15}$$

with $\phi = \mathrm{GELU}$ and $W_h \in \mathbb{R}^{d\times d}$ separate per horizon. The graph is applied once, at the output side, so Ψ_h keeps its meaning (source j at t informs target i at $t+h$) instead of being used as a recurrent smoother at every input step. The cost is a sparse product, $O(|E_h|\,d)$ per sample. A second hop, $S_h^{(2)} = \phi(\hat{G}_h S_h W_h')$, is an ablation, not the default (the $h = 12$ support already reaches hundreds of nodes).

B.4 Horizon-conditioned gate

A learned embedding $e_h \in \mathbb{R}^{d_e}$ per horizon (broadcast over nodes) conditions a node-wise gate that mixes the spatial and temporal representations:

$$\alpha_h = \sigma\Big(\mathrm{MLP}_g\big([\,Z \,\|\, S_h \,\|\, e_h\,]\big)\Big) \in (0,1)^{B\times N\times d}, \qquad H_h = \alpha_h \odot S_h + (1-\alpha_h)\odot Z. \tag{16}$$

MLP_g has one hidden layer of width d and is shared across horizons (the horizon enters through e_h); this keeps the parameter count identical whether one or three graphs are used. The gate can close the graph where it hurts, so the model is never forced to use neighbours. The horizon-wise mean gate $\bar{\alpha}_h = \mathbb{E}[\alpha_h]$ is logged: it is a direct readout of whether the network relies more on spatial information at longer horizons.

B.5 Direct multi-horizon heads

$$\hat{y}_h = H_h\, w_h^o + b_h^o \in \mathbb{R}^{B\times N}, \qquad h \in \mathcal{H}, \tag{17}$$

produced in one forward pass; no prediction is fed back as an input. Predictions are de-standardised with the training statistics before scoring.

B.6 Loss

$$\mathcal{L} = \frac{1}{|\mathcal{H}|}\sum_{h\in\mathcal{H}} \frac{1}{|M_h|} \sum_{(b,n)\in M_h} \big|\hat{y}_{h,b,n} - y_{h,b,n}\big|, \tag{18}$$

a masked mean absolute error with uniform horizon weights (horizon weighting is an ablation, not a default; the long-horizon gains reported in the literature came from architecture, not from loss weights).

B.7 Size and cost (design estimate, $d = 64$, $d_e = 8$, $F = 4$)

Component	Parameters	Compute per sample
GRU encoder (shared over nodes)	$3\,[d(F+d)+2d] \approx 13.4$k	$O(NTd^2)$
Propagation W_h, b_h (per horizon)	$d^2+d \approx 4.2$k each	$O(\lvert E_h\rvert\, d + Nd^2)$
Gate MLP (shared) + e_h	$(2d+d_e)d + d^2 + 2d \approx 13.0$ k	$O(Nd^2)$ per horizon
Heads (per horizon)	$d+1$	$O(Nd)$
Total, three horizons	≈ 39k	dominated by the GRU; the graph terms are sparse ($\lvert E_h\rvert \le 51$k)

For reference, the A3T-GCN baseline carries about 30k weights in its graph-GRU cell plus the attention and output layers, and performs a sparse $N \times N$ product at every one of the 12 input steps; HICP performs one per horizon. HICP is therefore lighter per step and fits the 4 GB-GPU, 4–6 hour Kaggle budget of the existing runs.

B.8 Experimental configurations (architecture fixed; only the adjacency tensors change)

Configuration	Adjacency in branch $h = 3$ / 6 / 12	What it tests
HICP family (proposed)	G_3 / G_6 / G_{12}	the full hypothesis
HICP G_{12} everywhere	G_{12} / G_{12} / G_{12}	does horizon matching matter beyond long reach? (**the decisive comparison**)
HICP G_3 everywhere	G_3 / G_3 / G_3	short reach at every horizon (reach control)
HICP shuffled assignment	e.g. G_{12} / G_3 / G_6	is the pairing graph ↔ horizon what matters?
HICP road graph / road power	A_{road}, or A_{road}^{k} with k matched to reach	do the VAR dynamics add anything over hop distance?
HICP BH-Granger	pairwise Granger F-tests with Benjamini–Hochberg (GC-MGT style), one graph	conditional knockoff selection vs marginal testing
HICP no graph	$S_h \equiv 0$, gate forced to $\alpha_h = 0$	per-node floor
A3T-GCN: no graph / G_{12} / G_h	as supported by the baseline loader	does the generated family help an existing architecture?

C. Training and evaluation protocol

- **Splits.** Chronological train / validation / test as in all earlier runs; graphs, scalers and every hyper-parameter choice (including q) use training and validation rows only; the test period is scored once per run.
- **Optimisation.** AdamW, learning rate 3×10^{-4} with linear warm-up (one epoch) and cosine decay, gradient-norm clipping at 5, at least 30 epochs with early stopping on validation MAE (patience 6). The same recipe is applied to every configuration, including the A3T-GCN baseline re-runs: the earlier

recipe (learning rate 10^{-3}, batch 8, 20 fixed epochs, no schedule) produced run-to-run swings of 2–3 RMSE, larger than the effects under study.

- **Seeds.** At least three seeds $\{42, 7, 123\}$ for every row; five for the decisive pair (family against G_{12} everywhere).
- **Metrics per horizon.** MAE, RMSE and masked MAPE (ground truth below 10 vehicles per 5 minutes excluded), each at $h = 3, 6, 12$ separately, plus their mean over $\mathcal{H}$ and the error growth $\mathrm{RMSE}(12)/\mathrm{RMSE}(3)$. Means and standard deviations over seeds, parameter counts and time per epoch are reported. Averages over three horizons are not comparable with published 12-step LargeST averages and are labelled accordingly.
- **Significance.** Within a seed: paired bootstrap over sensors of the RMSE ratio (2,000 resamples, 95% interval). Across seeds: sign consistency and the mean paired difference; with three to five seeds a paired t-test has little power and is reported but not relied on.
- **Decision rule, fixed in advance.** "Horizon-indexed propagation helps" is claimed only if the HICP family beats HICP-G_{12}-everywhere at $h = 6$ and $h = 12$ with the same sign in every seed and a within-seed interval excluding 1 in the majority of seeds. Otherwise the result is reported as "long-reach graph generation helps; horizon matching does not add to it."

D. Hypotheses and the experiments that decide them

#	Hypothesis	Comparison
H1	The generated family improves medium- and long-horizon accuracy over no graph	HICP family vs HICP no-graph, at $h = 6, 12$
H2	Horizon-indexed propagation beats a shared graph	HICP family vs G_{12} everywhere (primary), vs G_3 everywhere, vs shuffled
H3	The gain grows with horizon	relative improvement at $h = 3$ vs 6 vs 12; mean gate $\bar{\alpha}_h$ increasing in h
H4	The family beats a physical and a conventional Granger graph	HICP family vs road / road-power / BH-Granger
H5	Improvements are consistent across seeds	sign consistency over at least 3 seeds; five for H2

E. Open design decisions

1. q per horizon (0.2 or 0.3; chosen on validation; the power at $h = 12$ is unknown until the selection is run).
2. Weighted G_h (eq. 11) versus binary support in eq. 14; both will be run as an ablation.
3. Heads at $h \in \{3, 6, 12\}$ only (graphs exist there) versus all 12 steps (which would require a graph for every step or an interpolation rule).
4. A modern baseline under the same protocol (Graph WaveNet or GC-MGT) in addition to A3T-GCN.
5. Pooling of the encoder output: last state (eq. 13) versus attention over the T states.

Implementation note. Part A lives in the existing pipeline (restricted VAR, impulse responses, candidate pool, group-knockoff selection and graph assembly). Part B is to be written in PyTorch with the data windows, scaling, masks and metrics ported from the existing TensorFlow pipeline and checked for equality on a sample before any comparison with the A3T-GCN runs.